

Rajalakshmi Engineering College

Name: Sheik Abdulla
Email: 240701493@rajalakshmi.edu.in
Roll no: 240701493
Phone: 6379676782
Branch: REC
Department: CSE - Section 9
Batch: 2028
Degree: B.E - CSE

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 3_Q4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Sesha is developing a weather monitoring system for a region with multiple weather stations. Each weather station collects temperature data hourly and stores it in a 2D array.

Write a program that can add the temperature data from two different weather stations to create a combined temperature record for the region.

Input Format

The first line of input consists of two space-separated integers N and M, representing the number of rows and columns of the matrices, respectively.

The next N lines consist of M space-separated integers, representing the values of the first matrix.

The following N lines consist of M space-separated integers, representing the values of the second matrix.

Output Format

The output prints the addition of the two matrices in N rows and M columns, representing the combined temperature record.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3 3

1 2 3

4 5 6

7 8 9

1 1 1

2 2 2

3 3 3

Output: 2 3 4

6 7 8

10 11 12

Answer

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);
```

```
        // Read the number of rows (N) and columns (M)
```

```
        int N = sc.nextInt();
```

```
        int M = sc.nextInt();
```

```
        int[][] matrix1 = new int[N][M];
```

```
        int[][] matrix2 = new int[N][M];
```

```
        int[][] result = new int[N][M];
```

```
        // Read first matrix
```

```
        for (int i = 0; i < N; i++) {
```

```
            for (int j = 0; j < M; j++) {
```

```
        matrix1[i][j] = sc.nextInt();
    }
}

// Read second matrix
for (int i = 0; i < N; i++) {
    for (int j = 0; j < M; j++) {
        matrix2[i][j] = sc.nextInt();
    }
}

// Add matrices element-wise
for (int i = 0; i < N; i++) {
    for (int j = 0; j < M; j++) {
        result[i][j] = matrix1[i][j] + matrix2[i][j];
    }
}

// Print the result matrix
for (int i = 0; i < N; i++) {
    for (int j = 0; j < M; j++) {
        System.out.print(result[i][j] + " ");
    }
    System.out.println();
}

sc.close();
}
```

Status : Correct

Marks : 10/10